Marco Del Fava

Research Software & Neural Data
M.S. Biomedical Engineering student | UC Davis | Human intracranial electrophysiology
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Research Profile

Biomedical Engineering M.S. student at UC Davis developing Python tools for neuroscience data acquisition, conversion, and validation. I work with human intracranial recordings, including proprietary Neuralynx, Intan, and MED formats. My projects span reader development, metadata repair, signal-integrity testing, and agent-assisted software engineering.

Technical Skills

Scientific Programming: Python, NumPy, Pandas, scikit-learn, pytest, Git, Linux, Docker; MATLAB

Neural Data & Acquisition: Neuralynx, Intan RHD, DarkHorse MED, Neo, SpikeInterface, Kilosort4, MountainSort5

Engineering Methods: Scientific data parsing, metadata validation and repair, waveform analysis, experimental synchronization, profiling, regression tests

Research Experience

Graduate Student, UC Davis Health — Z-Lab (Dr. Jie Zheng) — Jul 2025 - Present

- Work with human intracranial electrophysiology recordings, developing Python workflows to prepare, process, and analyze single-neuron data.
- Develop event-aligned analyses of responses to repeated spoken narratives, combining sorted spike timestamps, experimental markers, rasters, and peri-stimulus firing rates.
- Support acquisition and spike-processing workflows across recording systems, including channel mapping, signal quality assessment, and synchronization with behavioral experiments.

Selected Engineering Projects

Neural Data Formats and Integrity

- Adapted a Neo-backed MED reader to skip corrupt parity channels whose metadata caused standard readers to abort, allowing processing without modifying source recordings.
- Developed validation and repair utilities for Neuralynx recordings with sample-count inconsistencies and missing recording-end metadata that prevented downstream conversion.
- Evaluated signal fidelity across 20 million samples on one channel (32 kHz; 625 s), comparing original CSC and compressed MED recordings. Measured a maximum amplitude difference of 0.5 μV, with exact agreement after whole-microvolt source rounding.

Neural Acquisition and Experimental Synchronization

- Built Python workflows for processing Intan recordings through SpikeInterface, including probe mapping, channel checks, spike sorting, waveform extraction, and quality metrics.
- Developed synchronization software to align PsychoPy behavioral events with RHX recordings, using matched event labels and affine clock fitting to characterize drift between recording systems.

Waveform Analysis and Spike Sorting

- Evaluated waveform representations across 38,145 spike snippets and 10 putative O-Sort unit labels, comparing how well different features preserved local waveform structure and supported clustering.
- Found that strong local neighbor consistency did not translate into equivalent global clustering agreement, distinguishing local waveform-feature consistency from whole-dataset clustering behavior.

Debugging Agent-Generated Software

- Investigated a performance regression in Hosaka, a command-line software platform, where agent-generated schema validation made listing seven records take approximately 70 seconds.
- Used execution timing logs to identify repeated validation as the bottleneck, then introduced content-based caching and regression tests to preserve checks for changed or invalid schemas.

Additional Professional Experience

IT Administrator, First Step Communities — Jul 2021 - Present

- Maintain network and IT infrastructure across multiple shelter sites, including routing/VLAN segmentation, Microsoft 365 administration, and troubleshooting.
- Support longitudinal data collection and analysis of service utilization and housing outcomes, translating operational data into research and reporting workflows.

Analyst, Eurofins Test America — Oct 2019 - Jun 2021

- Prepared soil and water samples using solvent extraction in a high-throughput environmental testing laboratory, following established quality assurance procedures.
- Maintained high-accuracy data entry and review in a regulated laboratory environment; supported QA documentation and adherence to standard operating procedures.

Education

M.S. Biomedical Engineering, University of California, Davis — 2026–Present

M.S. Psychology, Arizona State University — 2023

B.S. Neuroscience, Physiology, and Behavior, University of California, Davis — 2017

Professional Development & Certifications

Python for Data Science, Professional Concentration, UC Davis Continuing and Professional Education — 2025

- Data Analysis with Matplotlib, Pandas, and Scikit-Learn Python Libraries — UC Davis Continuing and Professional Education (2025-09-18)
- Biomedical Research; Research Administration; GCP Social and Behavioral Research — CITI Program